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Light-emitting module

Abstract

A light-emitting module including a first wiring substrate, a second wiring substrate located on the first wiring substrate, a plurality of light-emitting elements located on the second wiring substrate, a wire connecting a first terminal of the first wiring substrate and a second terminal of the second wiring substrate, a first resin located on the first wiring substrate and on the second wiring substrate, and a protective member located on the first resin. The first resin covers the wire and surrounds the plurality of light-emitting elements when viewed in plan view. The protective member includes a light-transmitting member covering the first resin when viewed in plan view, and a light-shielding film located at a surface of the light-transmitting member. The light-shielding film covers the first resin when viewed in plan view.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

(1) This application is based upon and claims the benefit of priority from the prior Japanese Patent Application No. 2021-175418, filed on Oct. 27, 2021; the entire contents of which are incorporated herein by reference.

BACKGROUND

(2) Embodiments of the disclosure relate to light-emitting modules.

(3) In recent years, light-emitting modules that can individually drive multiple light-emitting elements have been used as light sources of adaptive driving beams (ADB) and the like. ADBs are designed to project a desired light distribution pattern frontward by controlling the lighting pattern of the multiple light-emitting elements.

SUMMARY

(4) The light-emitting module may be damaged if external light such as sunlight or the like travels backward through the optical system and is concentrated at the light-emitting surface vicinity.

(5) According to one aspect of the present invention, a light-emitting module includes a first wiring substrate, a second wiring substrate located on the first wiring substrate, a plurality of light-emitting elements located on the second wiring substrate, a wire connecting a first terminal of the first wiring substrate and a second terminal of the second wiring substrate, a first resin located on the first wiring substrate and on the second wiring substrate, and a protective member located on the first resin. The first resin covers the wire and surrounding the plurality of light-emitting elements when viewed in plan view. The protective member includes a light-transmitting member covering the first resin when viewed in plan view, and a light-shielding film located at a surface of the light-transmitting member. The light-shielding film covers the first resin when viewed in plan view.

(6) According to an embodiment, a light emitting module capable of suppressing damage due to external light can be realized.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a perspective view showing a light-emitting module according to a first embodiment when viewed obliquely from above;

(2) FIG. 2 is a perspective view showing the light-emitting module according to the first embodiment when viewed obliquely from above;

(3) FIG. 3 is a perspective view showing the light-emitting module according to the first embodiment when viewed obliquely from below;

(4) FIG. 4 is an enlarged plan view showing region IV of FIG. 2;

(5) FIG. 5 is a cross-sectional view along line V-V shown in FIG. 2;

(6) FIG. 6 is a plan view showing the light-emitting module according to the first embodiment;

(7) FIG. 7 is a cross-sectional view showing a protective member according to the first embodiment;

(8) FIG. 8 is a cross-sectional view showing a protective member of a first modification of the first embodiment;

(9) FIG. 9 is a cross-sectional view showing a protective member of a second modification of the first embodiment;

(10) FIG. 10 is a cross-sectional view showing a protective member according to a second embodiment;

- (11) FIG. 11 is a plan view showing a light-emitting module according to a third embodiment;
(12) FIG. 12 is a side view showing the light-emitting module according to the third embodiment;
(13) FIGS. 13A to 13D are cross-sectional views showing a method for forming a ventilation portion according to the third embodiment; and
(14) FIGS. 14A to 14D are cross-sectional views showing a method for forming a ventilation portion of a modification of the third embodiment.

DETAILED DESCRIPTION

(15) Embodiments of the invention will now be described with reference to the drawings. The drawings are schematic and are enhanced or simplified as appropriate. For example, in some of the drawings, compared to the light-emitting module, the light-emitting elements are illustrated larger than actual sizes, and the protective members are illustrated thicker than actual thicknesses. The dimensional ratios of the components do not always match between the drawings. End views that show only cross sections may be used as cross-sectional views.

First Embodiment

(16) A light-emitting module **1** according to a first embodiment includes a first wiring substrate **10**, a second wiring substrate **20** located on the first wiring substrate **10**, multiple light-emitting elements **30** located on the second wiring substrate **20**, a wire **60** connecting a first terminal **12** of the first wiring substrate **10** and a second terminal **22** of the second wiring substrate, a first resin **70**, and a protective member **80** located on the first resin **70**. The first resin **70** is located on the first wiring substrate **10** and on the second wiring substrate **20**, covers the wire **60**, and surrounds the multiple light-emitting elements **30** when viewed in plan view. The protective member **80** includes a light-transmitting member **81** that covers the first resin **70** when viewed in plan view, and a light-shielding film **82** that is located at the surface of the light-transmitting member **81** and covers the first resin **70** when viewed in plan view.

(17) FIG. 1 is a perspective view showing the light-emitting module according to the embodiment when viewed obliquely from above.

(18) FIG. 2 is a perspective view showing the light-emitting module according to the embodiment when viewed obliquely from above.

(19) FIG. 3 is a perspective view showing the light-emitting module according to the embodiment when viewed obliquely from below.

(20) FIG. 4 is an enlarged plan view showing region IV of FIG. 2.

(21) FIG. 5 is a cross-sectional view along line V-V shown in FIG. 2.

(22) FIG. 6 is a plan view showing the light-emitting module according to the embodiment.

(23) FIG. 7 is a cross-sectional view showing the protective member according to the embodiment.

(24) The protective member **80** is not illustrated in FIGS. 2 to 4.

(25) In the specification, an XYZ orthogonal coordinate system is employed for convenience of description. The direction from the first wiring substrate **10** toward the protective member **80** is taken as a “Z-direction”. The longitudinal direction of the light-emitting module **1** is taken as an “X-direction”; and the transverse direction is taken as a “Y-direction”. Although the Z-direction also is called “up”, and the opposite direction of the Z-direction also is called “down”, these expressions are for convenience and are independent of the direction of gravity. In the specification, the expression “when viewed in plan view” means when viewed from the Z-direction.

(26) As shown in FIGS. 1 to 3, the light-emitting module **1** of the embodiment includes the first wiring substrate **10**, the second wiring substrate **20**, the multiple light-emitting elements **30**, a second resin **40**, a third resin **50**, the multiple wires **60**, the first resin **70**, and the protective member **80**. For convenience of illustration in FIG. 2, the entire protective member **80**, a portion of the first resin **70**, and a portion of the third resin **50** are not illustrated so that a portion of the wire **60**, a portion of the light-emitting element **30**, etc., are visible.

(27) The first wiring substrate **10** includes, for example, a flat-plate base body **11**, and wiring

located at least at the upper surface of the base body **11**. The first wiring substrate **10** can have a flat plate shape, or can have a cavity in the upper surface to house the second wiring substrate **20**. It is favorable for the material of the base body **11** to have good heat dissipation, and more favorable to have a high light-shielding property and strength. Specifically, examples of the material of the base body **11** include metals such as aluminum (Al), copper (Cu), and the like, ceramics such as aluminum oxide, aluminum nitride, silicon nitride, mullite, and the like, resins such as phenol, epoxy, polyimide, BT (bismaleimide triazine), polyphthalamide (PPA), and the like, a composite including a resin, a metal, or a ceramic, etc. Examples of the material of the wiring include, for example, metals such as copper (Cu), silver (Ag), gold (Au), aluminum (Al), platinum (Pt), titanium (Ti), tungsten (W), palladium (Pd), iron (Fe), nickel (Ni), and the like, alloys of such metals, etc.

(28) As an example, the first wiring substrate **10** includes a stack of metallic members of aluminum (Al), copper (Cu), or the like and an insulating member such as an epoxy resin or the like, and wiring located inside and at surfaces of the first wiring substrate **10**. A portion of the wiring forms the multiple first terminals **12** at an upper surface **10a** of the first wiring substrate **10**, and another portion of the wiring forms multiple lower surface pads **13** at a lower surface **10b** of the first wiring substrate **10**.

(29) A heat dissipation portion **14** is configured by exposing the metallic member from under the insulating member at the upper surface **10a** and the lower surface **10b** of the first wiring substrate **10**. When viewed in plan view, the heat dissipation portion **14** is located at the central portion of the first wiring substrate **10**, and the first terminals **12** and the lower surface pads **13** are located at two sides of the heat dissipation portion **14** with the heat dissipation portion **14** interposed. For example, the first terminals **12** and the lower surface pads **13** are arranged along the long sides of the first wiring substrate **10**.

(30) The second wiring substrate **20** is located on the heat dissipation portion **14** of the first wiring substrate **10**. The second wiring substrate **20** is, for example, a silicon substrate, e.g., an application specific integrated circuit substrate (ASIC substrate) having an embedded integrated circuit. The lower surface of the second wiring substrate **20** is bonded to the upper surface of the heat dissipation portion **14** via a bonding member. For example, a silver paste that includes a silicone resin is an example of the bonding member. The central portion of an upper surface **21** of the second wiring substrate **20** is a region where the light-emitting element **30** is placed. An upper surface pad that is connected to the light-emitting elements **30** is located in this region, and the second terminals **22** that are electrically connected with the upper surface pad are located in the peripheral portion of this region.

(31) The wire **60** is a member for electrically connecting the first wiring substrate **10** and the second wiring substrate **20**. The wire **60** is connected to the first terminal **12** of the first wiring substrate **10** and the second terminal **22** of the second wiring substrate **20**. For example, gold (Au) is an example of the material of the wire **60**. For example, the number of the wires **60** is equal to the number of the first terminals **12**. The multiple wires **60** are located along the outer edge of the second wiring substrate and straddle the outer edge of the second wiring substrate **20**.

(32) As shown in FIG. 2, FIG. 4, and FIG. 5, the multiple light-emitting elements **30** are placed on the central portion of the upper surface **21** of the second wiring substrate **20**. Each light-emitting element **30** is substantially rectangular when viewed in plan view. For example, the multiple light-emitting elements **30** are arranged in a matrix configuration. In an example, four segments each include 64 rows and 64 columns of the light-emitting elements **30** having substantially square upper surfaces, for a total of 16,384 light-emitting elements **30**. In an example, the arrangement interval of the light-emitting element **30** is 45 μm to 55 μm , and the length of one side of each light-emitting element **30** is 40 μm to 50 μm . Accordingly, the distance between adjacent light-emitting elements **30** is 5 μm to 15 μm . In an example, the thickness of the light-emitting element **30** is 5 μm to 10 μm . The light-emitting element **30** is connected to the upper surface pad of the

upper surface **21** of the second wiring substrate **20**. The light-emitting element **30** is, for example, a light-emitting diode (LED) that emits, for example, blue light.

(33) Elements that emit light of any wavelength can be selected as the light-emitting elements **30**. For example, a light-emitting element that includes ZnSe, a nitride semiconductor ($\text{In.sub.xAl.sub.yGa.sub.1-x-yN}$, $0 \leq x$, $0 \leq y$, and $x+y < 1$), or GaP can be selected as a light-emitting element emitting blue or green light. A semiconductor such as GaAlAs or AlInGaP can be favorably used as the light-emitting element **30** that emits red light. Furthermore, semiconductor light-emitting elements that are made of other materials also can be used. The composition and the light emission color of the light-emitting element **30** can be selected as appropriate according to the purpose.

(34) The second resin **40** is light-reflective. The second resin **40** is located between the adjacent light-emitting elements **30**. The second resin **40** also can be located between the upper surface **21** of the second wiring substrate **20** and the lower surfaces of the light-emitting elements **30**. That is, the second resin **40** can cover the side surface and the lower surface of the light-emitting element **30** and leave the upper surface of the light-emitting element **30** exposed. The second resin **40** can cover the entire side surface of the light-emitting element **30**, or can leave exposed an upper portion of the side surface linked to the upper surface **21** of the light-emitting element **30**. By covering the side surface of the light-emitting element **30** with a light-reflective second resin, more of the light emitted from the light-emitting element **30** can be extracted from the upper surface **21**. The second resin **40** includes a base material made of a light-transmitting resin, and a light-reflective substance included in the base material. The light extraction efficiency from the light-emitting element **30** can be increased by increasing the content of the light-reflective substance included in the second resin **40**. From the perspective of the increase of the light extraction efficiency and the fluidity of the resin, it is favorable for the concentration of the light-reflective substance in the second resin **40** to be not less than 50% by mass and not more than 70% by mass, e.g., about 60% by mass.

(35) For example, a resin such as silicone, modified silicone, epoxy, modified epoxy, acrylic, or a hybrid resin that includes at least one of these resins can be used as the light-transmitting resin included in the base material of the second resin **40**. Among these resins, silicone that has excellent heat resistance and lightfastness is favorable, and a dimethyl silicone resin is more favorable. Dimethyl silicone resins can be favorably used as materials of automotive applications due to their excellent reliability such as high-temperature tolerance, etc.

(36) For example, titanium oxide, aluminum oxide, zinc oxide, barium carbonate, barium sulfate, boron nitride, aluminum nitride, a glass filler, etc., can be favorably used as the light-reflective substance.

(37) As an example, the base material is a dimethyl silicone resin; and the light-reflective substance is titanium oxide. The exterior color of the second resin **40** is white.

(38) The third resin **50** is transmissive and covers the upper surfaces of the light-emitting elements **30** and the upper surface of the second resin **40**. The third resin **50** contacts the upper surfaces of the light-emitting elements **30**, the upper portions of the side surfaces of the light-emitting elements **30**, and the upper surface of the second resin **40**. The third resin **50** includes at least a base material made of a light-transmitting resin; and a fluorester can be included in the base material. In an example, the thickness of the third resin **50** is 25 μm to 35 μm .

(39) Materials similar to the base material of the second resin **40** described above can be used as the base material of the third resin **50**. As the fluorester of the third resin **50**, an yttrium-aluminum-garnet-based fluorester (e.g., $\text{Y.sub.3(Al, Ga).sub.5O.sub.12:Ce}$), a lutetium-aluminum-garnet-based fluorester (e.g., $\text{Lu.sub.3(Al, Ga).sub.5O.sub.12:Ce}$), a terbium-aluminum-garnet-based fluorester (e.g., $\text{Tb.sub.3(Al, Ga).sub.5O.sub.12:Ce}$), a CCA-based fluorester (e.g., $\text{Ca.sub.10(PO.sub.4).sub.6Cl.sub.2:Eu}$), an SAE-based fluorester (e.g., $\text{Sr.sub.4Al.sub.14O.sub.25:Eu}$), a chlorosilicate-based fluorester (e.g., $\text{Ca.sub.8MgSi.sub.4O.sub.16Cl.sub.2:Eu}$), an oxynitride-based fluorester such as β -sialon-based

(e.g., (Si, Al).sub.3(O, N).sub.4:Eu) or α -sialon-based (e.g., Ca(Si, Al).sub.12(O, N).sub.16:Eu), an SLA-based fluorescer (e.g., SrLiAl.sub.3N.sub.4:Eu), a nitride-based fluorescer such as CASN-based (e.g., CaAlSiN.sub.3:Eu) or SCASN-based (e.g., (Sr, Ca)AlSiN.sub.3:Eu), a KSF-based fluorescer (e.g., K.sub.2SiF.sub.6:Mn), a fluoride-based fluorescer such as KSAF-based (e.g., K.sub.2Si.sub.0.99Al.sub.0.01F.sub.5.99:Mn) or MGF-based (e.g., 3.5MgO.0.5MgF.sub.2.Math.GeO.sub.2:Mn), a fluorescer having a perovskite structure (e.g., CsPb(F, Cl, Br, I).sub.3), a quantum dot fluorescer (e.g., CdSe, InP, AgInS.sub.2, or AgInSe.sub.2), etc., can be used.

(40) The KSAF-based fluorescer can include the composition of the following formula (I).



(41) In formula (I), M is an alkaline metal and can include at least K. Mn can be a tetravalent Mn ion. p, q, r, and s can satisfy $0.9 \leq p+q+r \leq 1.1$, $0 < q \leq 0.1$, $0 < r \leq 0.2$, and $5.9 \leq s \leq 6.1$. It is favorable to be $0.95 \leq p+q+r \leq 1.05$ or $0.97 \leq p+q+r \leq 1.03$, $0 < q \leq 0.03$, $0.002 \leq r \leq 0.02$, or $0.003 \leq q \leq 0.015$, $0.005 \leq r \leq 0.15$, $0.01 \leq r \leq 0.12$, or $0.015 \leq r \leq 0.1$, and $5.92 \leq s \leq 6.05$ or $5.95 \leq s \leq 6.025$. For example, compositions of K.sub.2[Si.sub.0.946Al.sub.0.005Mn.sub.0.049F.sub.5.995], K.sub.2[Si.sub.0.942Al.sub.0.008Mn.sub.0.050F.sub.5.992], and K.sub.2[Si.sub.0.939Al.sub.0.014Mn.sub.0.047F.sub.5.986] are examples. According to such a KSAF-based fluorescer, a red light emission that has high luminance and a narrow width at half maximum at the light emission peak wavelength can be obtained.

(42) The first resin **70** protects the wires **60** connecting the first wiring substrate **10** and the second wiring substrate **20**. When viewed in plan view, the first resin **70** has a rectangular frame shape along the outer edge of the second wiring substrate **20**. The first resin **70** is located over the upper surface of the second wiring substrate **20** from the upper surface of the first wiring substrate **10** and covers the first terminals **12** of the first wiring substrate **10**, the wires **60**, and the second terminals **22** of the second wiring substrate **20**. In an example, the height from the upper surface of the second wiring substrate **20** to the apex of the first resin **70** is 200 μm to 500 μm . Here, the apex of the first resin means the highest position of the first resin in a cross section passing through a light-emitting surface **38**.

(43) The first resin **70** can be transmissive or light-shielding. It is favorable for the first resin **70** to be light-shielding, and more favorably light-absorbing. Stray light caused by the first resin **70** reflecting light emitted from the light-emitting elements **30** can be suppressed thereby. For example, the first resin **70** is black or gray. The first resin **70** includes at least a base material made of a light-transmitting resin, and can include a light-reflective substance and/or a light-absorbing substance in the base material. Materials similar to the base material of the second resin **40** described above can be used as the base material. Materials similar to the light-reflective substance of the second resin **40** described above can be used as the light-reflective substance. A black pigment such as carbon black or the like, graphite, etc., are examples of the light-absorbing substance.

(44) As shown in FIG. 5, the first resin **70** includes an outer resin frame **71** located on the first wiring substrate **10**, an inner resin frame **72** located on the second wiring substrate **20**, and a protective resin **73** located between the outer resin frame **71** and the inner resin frame **72**. The wires **60** are located inside the protective resin **73**. According to the embodiment, for example, the first resin **70** includes the outer resin frame **71**, the inner resin frame **72**, and the protective resin **73**. The protective resin **73** protects the wires; and the outer resin frame **71** and the inner resin frame **72** hold the protective resin **73**. The outer resin frame **71** and the inner resin frame **72** are located respectively on the first wiring substrate **10** and on the second wiring substrate **20** with the wires interposed. The protective resin **73** is located in the region between the outer resin frame **71** and the inner resin frame **72**. As an example, the outer resin frame **71**, the inner resin frame **72**, and the protective resin **73** are light-shielding and have, for example, a black appearance. As necessary, the second resin **40**, the third resin **50**, and the first resin **70** can have different exterior colors and can

include colorants, light-diffusing materials, fillers for adjusting the viscosity, etc.

(45) The region of the upper surface of the third resin **50** surrounded with the first resin **70** is used as the light-emitting surface **38** of the light-emitting module **1**. In other words, the first resin **70** surrounds the light-emitting surface **38**. The multiple light-emitting elements **30** and the second resin **40** are located under the third resin **50** at the light-emitting surface **38**. The height from the upper surface of the second wiring substrate **20** to the upper surface of the light-emitting surface **38** is less than the height from the upper surface of the second wiring substrate **20** to the apex of the first resin **70**. That is, the light-emitting module **1** includes a recess surrounded with the first resin **70** at the upper surface of the light-emitting module **1**. The upper surface of the third resin **50** is exposed from under the first resin **70** at the bottom of the recess and is included in the light-emitting surface **38** of the light-emitting module **1**. The light that is emitted from the multiple light-emitting elements **30** is reflected by the second resin **40**, enters the third resin **50**, and is emitted from the light-emitting surface **38**. A portion of the light emitted from the light-emitting surface **38** is reflected and/or absorbed by the first resin **70**.

(46) The protective member **80** is located on the first resin **70** and bonded to the first resin **70**. The protective member **80** includes the light-transmitting member **81**, and the light-shielding film **82** located at the surface of the light-transmitting member **81**. The light-shielding film **82** has a frame shape at the surface of the light-transmitting member **81** and covers the frame-shaped first resin **70**. The light-transmitting member **81** can have a frame shape similar to the light-shielding film **82**, or can have a plate shape covering the light-emitting surface **38** and the first resin **70**. The light-transmitting member **81** is made of a light-transmitting material; for example, glass or a resin can be used. Among light-transmitting materials, it is favorable to use glass that is not easily degraded by heat, light from the light-emitting surface, etc. For example, borosilicate glass and quartz glass are examples of the glass. The light-transmitting member **81** has, for example, a rectangular plate shape. The light-transmitting member **81** is, for example, a glass plate. The light-transmitting member **81** covers the first resin **70** when viewed in plan view. It is also favorable for the light-transmitting member **81** to cover the light-emitting surface **38** as well as the first resin **70**. In other words, it is favorable for the light-transmitting member **81** to cover the first resin **70** and cover the multiple light-emitting elements **30** when viewed in plan view. The adhesion of external dust, etc., to the light-emitting surface **38** can be suppressed thereby. The protective member **80** acts as a lid for the recess that is surrounded with the first resin **70**, thereby forming a space **75** on a light-emitting surface **32** that is shielded from the outside. The space **75** is, for example, an air layer located between the light-emitting surface **32** and the light-transmitting member **81**.

(47) The light-transmitting member **81** includes a lower surface **81a** and an upper surface **81b**. The lower surface **81a** faces the first resin **70**. The upper surface **81b** is at the side opposite to the lower surface **81a** and forms a portion of the upper surface of the light-emitting module **1**. It is favorable for the thickness of the light-transmitting member **81** to be not less than 0.2 mm and not more than 5.0 mm. By setting the thickness of the light-transmitting member **81** to be not less than 0.2 mm, the light-transmitting member **81** can have sufficient strength. By setting the thickness of the light-transmitting member **81** to be not more than 5.0 mm, the light-emitting module **1** can be thinned. It is more favorable for the thickness of the light-transmitting member **81** to be not less than 1.0 mm and not more than 3.0 mm. By setting the thickness of the light-transmitting member **81** to be not less than 1.0 mm, even when foreign matter adheres to the upper surface **81b** of the light-transmitting member **81**, the position of the foreign matter can be distant to the light-emitting surface **38**, i.e., the focal position of the optical system; therefore, foreign matter can be prevented from affecting the projection image. By setting the thickness of the light-transmitting member **81** to be not more than 3.0 mm, the light-emitting module **1** can be thinned even further.

(48) When viewed in plan view, the light-shielding film **82** has a frame shape that covers the first resin **70** but does not cover the multiple light-emitting elements **30**. When viewed in plan view, an inner edge **82i** of the light-shielding film **82** is located further inward than an inner edge **70i** of the

first resin **70**. It is favorable for the light-shielding film **82** to jut sufficiently inward with respect to the inner edge **70i** of the first resin **70** so that concentrated external light does not reach the first resin **70**. It is favorable for the light-shielding film **82** to be located at the lower surface **81a** of the light-transmitting member **81**, i.e., the surface at the first resin **70** side. The distance between the light-shielding film **82** and the first resin **70** can be reduced by providing the light-shielding film **82** at the lower surface **81a** of the light-transmitting member **81**; therefore, the inward jutting of the light-shielding film **82** described above can be reduced. That is, the light extraction efficiency can be increased because the surface area of the light-emitting surface **32** exposed from under the light-shielding film **82** when viewed in plan view can be increased. In an example, when the width of the first resin **70** when viewed in plan view is 260 μm , it is favorable for the jutting amount of the light-shielding film **82**, i.e., the minimum distance between the inner edge **82i** and the inner edge **70i** when viewed in plan view, to be not less than 200 μm . In such a case, it is favorable for all of the multiple light-emitting elements **30** to be separated from the light-shielding film **82** when viewed in plan view.

(49) As shown in JIG. 7, the light-shielding film **82** includes a lower layer **83** positioned at the first resin **70** side, and an upper layer **84** located higher than (i.e., at the light-transmitting member **81** side of) the lower layer **83**. The lower layer **83** has a higher optical absorptance than the upper layer **84**. In the specification, “optical absorptance” is the absorptance for the light emitted by the light-emitting elements **30**. By setting the optical absorptance of the lower layer **83** to be high, the light that is emitted from the light-emitting elements **30** can be effectively absorbed by the lower layer **83**, and stray light can be suppressed.

(50) On the other hand, the upper layer **84** has a higher light reflectance than the lower layer **83**. In the specification, “light reflectance” refers to the light reflectance to external light such as sunlight, refers to the reflectance to, for example, visible light, and refers to the reflectance to, for example, light of a wavelength of 400 nm. For example, the upper layer **84** has a reflectance of not less than 75% to light of a wavelength of 400 nm. Thereby, external light such as sunlight or the like is effectively reflected by the upper layer **84** of the light-shielding film **82** even when concentrated at the light-emitting surface vicinity of the light-emitting module **1**; therefore, the concentration at the first resin **70** and the resulting damage of the resin can be suppressed.

(51) For example, the lower layer **83** includes a chromium oxide layer, and, for example, a layer in which a chromium oxide layer **83a** and a chrome layer **83b** are stacked. The chromium oxide layer **83a** is made of chromium oxide (Cr.sub.2O.sub.3) and has a black exterior color. The chrome layer **83b** is made of chrome and has an appearance with metallic luster. For example, the upper layer **84** includes at least one type of metal selected from the group consisting of aluminum (Al), silver (Ag), and gold (Au) and is, for example, a layer in which an aluminum oxide layer **84a** and an aluminum layer **84b** are stacked. The aluminum oxide layer **84a** is made of aluminum oxide (Al.sub.2O.sub.3); and the aluminum layer **84b** is made of aluminum. In such a case, in the light-shielding film **82**, the aluminum layer **84b**, the alumina layer **84a**, the chrome layer **83b**, and the chromium oxide layer **83a** are stacked in this order downward from the light-transmitting member **81** side, i.e., toward the first resin **70** side. The lower layer **83** can be formed of a resin material that includes a light-absorbing substance.

(52) Effects of the embodiment will now be described.

(53) The light-emitting module **1** according to the embodiment includes the protective member **80** that is located on the first resin **70**. The protective member **80** includes the light-transmitting member **81** that covers the first resin, and the light-shielding film **82** that is located at the surface of the light-transmitting member **81** and covers the first resin **70**. Thereby, when the light-emitting module **1** is used as the light source of a headlamp of a vehicle, even when external light such as sunlight or the like travels backward through the optical system and is concentrated at the light-emitting module **1** vicinity, the damage of the first resin **70** due to the external light can be suppressed because the light-shielding film **82** shields the external light.

(54) Because the first resin **70** is black, the reflection by the first resin **70** of the light emitted from the light-emitting elements **30** can be suppressed, and the occurrence of stray light is suppressed. If the light-shielding film **82** is not located at the surface of the light-transmitting member **81**, there is a possibility that the first resin **70** may be burned when sunlight is concentrated at the black first resin **70**; however, according to the embodiment, burning of the first resin **70** can be suppressed because the protective member **80** includes the light-shielding film **82**.

(55) The light-shielding film **82** can suppress the degradation of the first resin **70** by external light because the upper layer **84** has high light reflectance and can effectively reflect the external light. The first resin **70** that protects the wires can be effectively protected thereby, which increases the reliability of the light-emitting module. Also, by setting the optical absorptance of the lower layer **83** to be high, the light that is emitted from the light-emitting elements **30** can be effectively absorbed, and the occurrence of stray light can be effectively suppressed. Thereby, a highly-accurate projection is possible when using the light-emitting module **1** as the light source of a headlamp of a vehicle. Thus, both protection from external light and suppression of stray light can be realized by the light-shielding film **82** having a stacked structure including the upper layer **84** that has a higher light reflectance and the lower layer **83** that has a higher optical absorptance.

(56) According to the embodiment, the inner edge **82i** of the light-shielding film **82** is located further inward than the inner edge **70i** of the first resin **70** when viewed in plan view. It is therefore more difficult for external light to reach the first resin **70**; and the first resin **70** can be more reliably protected from the external light.

(57) According to the embodiment, the light-shielding film **82** is located at the lower surface **81a** of the light-transmitting member **81**. The distance between the light-shielding film **82** and the first resin **70** in the Z-direction can be reduced thereby, and the first resin **70** can be sufficiently protected even without providing the inner edge **82i** of the light-shielding film **82** excessively inward with respect to the inner edge **70i** of the first resin **70**. As a result, the shielding by the light-shielding film **82** of the light emitted from the light-emitting elements **30** can be suppressed, and the light extraction efficiency of the light-emitting module **1** is increased.

(58) Because the light-transmitting member **81** covers the multiple light-emitting elements **30**, the light-emitting elements **30** can be protected from moisture, foreign matter flying in from the outside, etc.

(59) First Modification of First Embodiment

(60) FIG. **8** is a cross-sectional view showing a protective member of a first modification of the first embodiment.

(61) As shown in FIG. **8**, the light-emitting module according to the modification differs from the first embodiment in that the protective member **80** includes an anti-reflection coating **85** on at least one of the lower surface **81a** or the upper surface **81b** of the light-transmitting member **81**.

(62) In the example shown in FIG. **8**, the anti-reflection coating **85** of the protective member **80** is located at both the lower surface **81a** and the upper surface **81b** of the light-transmitting member **81**. The light-shielding film **82** is located under the anti-reflection coating **85** located at the lower surface **81a** of the light-transmitting member **81**. In other words, the anti-reflection coating **85** that is located at the lower surface **81a** of the light-transmitting member **81** is located between the light-shielding film **82** and the lower surface **81a** of the light-transmitting member **81**. Transmissive films of silicon oxide (SiO.sub.2), zirconium oxide (ZrO.sub.2), etc., can be used as single layers or a multilayer film as the anti-reflection coating **85**.

(63) According to the modification, because the protective member **80** includes the anti-reflection coating **85**, the reflection by the light-transmitting member **81** of the light emitted from the light-emitting elements **30** can be suppressed, and the light extraction efficiency of the light-emitting module can be increased. Otherwise, the configuration and the effects of the modification are similar to those of the first embodiment.

(64) Second Modification of First Embodiment

(65) FIG. 9 is a cross-sectional view showing a protective member of a second modification of the first embodiment.

(66) As shown in FIG. 9, the light-emitting module according to the modification differs from the first modification in that the anti-reflection coating **85** of the protective member **80** covers the light-shielding film **82**. In other words, in the protective member **80**, the light-shielding film **82** is located between the lower surface **81a** of the light-transmitting member **81** and the anti-reflection coating **85** located at the lower surface **81a** side of the light-transmitting member **81**. Otherwise, the configuration and the effects of the embodiment are similar to those of the first modification.

Second Embodiment

(67) FIG. 10 is a cross-sectional view showing a protective member according to a second embodiment.

(68) According to the embodiment as shown in FIG. 10, the protective member **80** includes the light-shielding film **82** at the upper surface **81b** of the light-transmitting member **81**. The light-shielding film **82** includes the upper layer **84** and the lower layer **83**, and the lower layer **83** is located further toward the first resin **70** side than the upper layer **84**, that is, at the light-transmitting member **81** side of the upper layer **84**. The light-shielding film **82** includes, for example, the chromium oxide layer **83a**, the chrome layer **83b**, the alumina layer **84a**, and the aluminum layer **84b** stacked in this order upward from the light-transmitting member **81** side, that is, in the direction away from the first resin **70**.

(69) Otherwise, the configuration according to the embodiment is similar to that of the first embodiment. According to the embodiment as well, the light that is emitted from the light-emitting elements **30** can be effectively absorbed, and the occurrence of stray light can be effectively suppressed. According to the embodiment as well, similarly to the first or second modification of the first embodiment, the protective member **80** can include the anti-reflection coating **85** on at least one of the lower surface **81a** or the upper surface **81b** of the light-transmitting member **81**.

Third Embodiment

(70) FIG. 11 is a plan view showing a light-emitting module according to a third embodiment.

(71) FIG. 12 is a side view showing the light-emitting module according to the embodiment.

(72) As shown in FIGS. 11 and 12, the light-emitting module **3** according to the embodiment includes a ventilation portion **90** between the first resin **70** and the protective member **80**. The ventilation portion **90** is a gap that links the space **75** surrounded with the first resin **70** to the outside. The ventilation portion **90** is, for example, a trench located at an upper surface **70a** of the first resin **70**. The upper surface **70a** of the first resin **70** includes a bonding region **76** that is bonded with the protective member **80**, and a non-bonding region **77** that is not bonded with the protective member **80**. In the light-emitting module **3**, the bonding region **76** is located higher than the non-bonding region **77**, and the ventilation portion **90** is formed between the non-bonding region **77** and the protective member **80**. When viewed in plan view, for example, the ventilation portion **90** is located at short-side portions of the rectangular frame-shaped first resin **70**. In other words, the ventilation portion **90** extends in the X-direction at the two Y-direction end portions of the first resin.

(73) A method for forming the ventilation portion **90** will now be described.

(74) FIGS. 13A to 13D are cross-sectional views showing the method for forming the ventilation portion according to the embodiment.

(75) As shown in FIG. 13A, the outer resin frame **71** is formed on the first wiring substrate **10**; and the frame-shaped inner resin frame **72** is formed on the second wiring substrate **20**. Then, an uncured resin material is disposed between the outer resin frame **71** and the inner resin frame **72** and cured. A lower layer resin layer **73a** is formed thereby.

(76) Then, as shown in FIG. 13B, an uncured resin material **73b** is selectively disposed on the lower layer resin layer **73a**. For example, the resin material **73b** is dispensed from a dispenser in paste form. At this time, the resin material **73b** is not located on the region at which the ventilation

portion **90** is to be formed.

(77) Continuing as shown in FIG. **13C**, the protective member **80** is disposed on the resin material **73b**.

(78) Then, as shown in FIG. **13D**, the resin material **73b** is cured while in contact with the protective member **80**. An upper layer resin layer **73c** is formed thereby. The upper layer resin layer **73c** is bonded to the protective member **80**.

(79) The protective resin **73** is formed of the lower layer resin layer **73a** and the upper layer resin layer **73c**. The first resin **70** is formed of the outer resin frame **71**, the inner resin frame **72**, and the protective resin **73**. At this time, the region at which the upper layer resin layer **73c** is not located on the lower layer resin layer **73a** is the ventilation portion **90**. Also, the region of the upper surface of the lower layer resin layer **73a** that is not covered with the upper layer resin layer **73c** is the non-bonding region **77**, and the region of the upper surface of the upper layer resin layer **73c** that contacts the protective member **80** is the bonding region **76**. In other words, the non-bonding region **77** is cured before placing the protective member **80**, and the bonding region **76** is cured after placing the protective member **80**.

(80) Effects of the embodiment will now be described.

(81) According to the embodiment, by providing the ventilation portion **90**, the space **75** that is surrounded with the first resin **70** communicates with the outside. Therefore, when manufacturing or using the light-emitting module **3**, the pressure inside the space **75** is not different from the outside pressure; and a force that is caused by the air pressure difference between the space **75** and the outside is not applied to the first resin **70** and the protective member **80**. As a result, the reliability of the light-emitting module **3** is increased.

(82) By providing the ventilation portion **90** in the short-side portion of the first resin **70**, the wires **60** are not exposed at the ventilation portion **90**. However, the ventilation portion **90** can be located at the long-side portion of the first resin **70** as long as there is no risk of exposing the wires **60**. Otherwise, the configuration and the effects according to the embodiment are similar to those of the first embodiment.

(83) Modification of Third Embodiment

(84) A method for forming the ventilation portion of a modification of the third embodiment is different from that of the third embodiment.

(85) FIGS. **14A** to **14D** are cross-sectional views showing the method for forming the ventilation portion of the modification.

(86) First, similarly to the third embodiment, the outer resin frame **71** is formed on the first wiring substrate **10** as shown in FIG. **13A**, and the frame-shaped inner resin frame **72** is formed on the second wiring substrate **20**. Then, the lower layer resin layer **73a** is formed by disposing an uncured resin material between the outer resin frame **71** and the inner resin frame **72** and by curing.

(87) Then, as shown in FIG. **14A**, a notch **78** is formed in the structure body made of the outer resin frame **71**, the inner resin frame **72**, and the lower layer resin layer **73a**. The notch **78** forms the ventilation portion **90** when viewed in plan view.

(88) Continuing as shown in FIG. **14B**, the uncured resin material **73b** is disposed on the lower layer resin layer **73a**. At this time, a trench **79** that reflects the notch **78** is formed in the upper surface of the resin material **73b**.

(89) Then, as shown in FIG. **14C**, the protective member **80** is disposed on the resin material **73b**.

(90) Continuing as shown in FIG. **14D**, the resin material **73b** is cured. The upper layer resin layer **73c** is formed thereby. The upper layer resin layer **73c** is bonded to the protective member **80**. The trench **79** becomes the ventilation portion **90** at this time.

(91) According to the modification, the position and size of the ventilation portion **90** can be controlled by the position and size of the notch **78**. The position and size of the ventilation portion **90** can be easily controlled thereby. Otherwise, the configuration, the formation method, and the effects of the modification are similar to those of the third embodiment.

- (92) Embodiments and their modifications described above are examples embodying the invention; and the invention is not limited to these embodiments and their modifications. For example, additions, deletions, or modifications of some of the components or processes of embodiments and their modifications described above also are included in the invention. Embodiments and their modifications described above also can be implemented in combination with each other.
- (93) For example, the invention can be utilized as a light source of a vehicle headlight, a display device, etc.
- (94) Embodiments include the following aspects.
- (95) Note 1. A light-emitting module, comprising: a first wiring substrate; a second wiring substrate located on the first wiring substrate; a plurality of light-emitting elements located on the second wiring substrate; a wire connecting a first terminal of the first wiring substrate and a second terminal of the second wiring substrate; a first resin located on the first wiring substrate and on the second wiring substrate, the first resin covering the wire and surrounding the plurality of light-emitting elements when viewed in plan view; and a protective member located on the first resin, the protective member including a light-transmitting member covering the first resin when viewed in plan view, and a light-shielding film located at a surface of the light-transmitting member, the light-shielding film covering the first resin when viewed in plan view.
- (96) Note 2. The module according to note 1, wherein the light-transmitting member covers the plurality of light-emitting elements when viewed in plan view.
- (97) Note 3. The module according to notes 1 or 2, wherein the light-transmitting member is made of glass.
- (98) Note 4. The module according to any one of notes 1 to 3, wherein a thickness of the light-transmitting member is not less than 0.2 mm and not more than 5.0 mm.
- (99) Note 5. The module according to any one of notes 1-4, wherein the light-shielding film is located at a surface of the light-transmitting member at the first resin side.
- (100) Note 6. The module according to any one of notes 1-5, wherein the light-shielding film has a frame shape when viewed in plan view, and an inner edge of the light-shielding film is located further inward than an inner edge of the first resin when viewed in plan view.
- (101) Note 7. The module according to any one of notes 1-6, wherein the light-shielding film includes: a lower layer positioned at the first resin side; and an upper layer located higher than the lower layer, the upper layer having a higher light reflectance than the lower layer.
- (102) Note 8. The module according to note 7, wherein the lower layer has a higher optical absorptance than the upper layer.
- (103) Note 9. The module according to any one of notes 1 to 6, wherein the light-shielding film includes: a lower layer positioned at the first resin side, the lower layer including chrome; and an upper layer located higher than the lower layer, the upper layer including aluminum.
- (104) Note 10. The module according to any one of notes 1 to 9, wherein an upper surface of the first resin includes: a bonding region bonded with the protective member; and a non-bonding region not bonded with the protective member.
- (105) Note 11. The module according to any one of notes 1 to 10, wherein the module has a ventilation portion between the first resin and the protective member, the ventilation portion links a space surrounded with the first resin to a space outside the module.
- (106) Note 12. The module according to note 11, wherein the ventilation portion is a trench located in an upper surface of the first resin.
- (107) Note 13. The module according to any one of notes 1 to 12, wherein the first resin is black.

Claims

1. A light-emitting module comprising: a first wiring substrate; a second wiring substrate located on the first wiring substrate; a plurality of light-emitting elements located on the second wiring

substrate; a wire connecting a first terminal of the first wiring substrate and a second terminal of the second wiring substrate; a first resin located on the first wiring substrate and on the second wiring substrate, the first resin covering the wire and surrounding the plurality of light-emitting elements when viewed in plan view; and a protective member located on the first resin, wherein the protective member includes: a light-transmitting member covering the first resin when viewed in plan view; and a light-shielding film located at a surface of the light-transmitting member, the light-shielding film covering the first resin when viewed in plan view, the light-shielding film includes: a lower layer positioned at a first resin side of the light-shielding film; and an upper layer located higher than the lower layer, and the upper layer has a higher light reflectance than the lower layer.

2. The module according to claim 1, wherein the light-transmitting member covers the plurality of light-emitting elements when viewed in plan view.

3. The module according to claim 1, wherein the light-transmitting member is glass.

4. The module according to claim 1, wherein a thickness of the light-transmitting member is not less than 0.2 mm and not more than 5.0 mm.

5. The module according to claim 1, wherein the light-shielding film is located at the surface of the light-transmitting member at a first resin side of the light transmitting member.

6. The module according to claim 1, wherein the light-shielding film has a frame shape when viewed in plan view, and an inner edge of the light-shielding film is located further inward than an inner edge of the first resin when viewed in plan view.

7. The module according to claim 1, wherein the lower layer has a higher optical absorptance than the upper layer.

8. A light-emitting module comprising: a first wiring substrate; a second wiring substrate located on the first wiring substrate; a plurality of light-emitting elements located on the second wiring substrate; a wire connecting a first terminal of the first wiring substrate and a second terminal of the second wiring substrate; a first resin located on the first wiring substrate and on the second wiring substrate, the first resin covering the wire and surrounding the plurality of light-emitting elements when viewed in plan view; and a protective member located on the first resin, wherein the protective member includes: a light-transmitting member covering the first resin when viewed in plan view; and a light-shielding film located at a surface of the light-transmitting member, the light-shielding film covering the first resin when viewed in plan view, and the light-shielding film includes: a lower layer positioned at a first resin side of the light-shielding film, the lower layer including chrome; and an upper layer located higher than the lower layer, the upper layer including aluminum.

9. The module according to claim 1, wherein an upper surface of the first resin includes: a bonding region bonded with the protective member; and a non-bonding region not bonded with the protective member.

10. The module according to claim 1, wherein the module has a ventilation portion between the first resin and the protective member, and the ventilation portion links a space surrounded with the first resin to a space outside the module.

11. The module according to claim 10, wherein the ventilation portion is a trench located in an upper surface of the first resin.

12. The module according to claim 1, wherein the first resin is black.

13. The module according to claim 8, wherein the module has a ventilation portion between the first resin and the protective member, and the ventilation portion links a space surrounded with the first resin to a space outside the module.

14. A light-emitting module comprising: a first wiring substrate; a second wiring substrate located on the first wiring substrate; a plurality of light-emitting elements located on the second wiring substrate; a wire connecting a first terminal of the first wiring substrate and a second terminal of the second wiring substrate; a first resin located on the first wiring substrate and on the second

wiring substrate, the first resin covering the wire and surrounding the plurality of light-emitting elements when viewed in plan view; and a protective member located on the first resin, wherein the protective member includes: a light-transmitting member covering the first resin when viewed in plan view; and a light-shielding film located at a surface of the light-transmitting member, the light-shielding film covering the first resin when viewed in plan view, the module has a ventilation portion between the first resin and the protective member, and the ventilation portion links a space surrounded with the first resin to a space outside the module.

15. The module according to claim 14, wherein the light-shielding film includes: a lower layer positioned at a first resin side of the light-shielding film; and an upper layer located higher than the lower layer, and the upper layer having a higher light reflectance than the lower layer.

16. The module according to claim 15, wherein the lower layer has a higher optical absorptance than the upper layer.

17. The module according to claim 14, wherein the light-shielding film includes: a lower layer positioned at a first resin side of the light-shielding film, the lower layer including chrome; and an upper layer located higher than the lower layer, the upper layer including aluminum.
